ISOLAR-A/B V12.7.0

What's New

ISOLAR-A V12.7.0 What's New

Overview

1. **Exclusive Area Lock Analysis***
2. **Secured PDU support in DBC Import***
3. **Merge Mode Support in Java Validation**
4. **File Level Validation Support for Custom Profile**
5. **General improvements**

*) New Feature in ISOLAR-A V12.7.0

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ISOLAR-A V12.7.0 What's New

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ISOLAR-A V12.7.0 What's New

Exclusive Area Lock Analysis (1/3)

Background Information

- Exclusive Area editor supports manual configuration of implementation mechanism for an exclusive area; however, this assignment is done based on user knowledge of best possible lock.

Feature Summary

- Exclusive Area editor is integrated with lock analysis functionality which parses all the c & h files from the project and proposes the best possible implementation mechanism type for exclusive area.
- This feature helps to optimize performance and avoid unnecessary locks.

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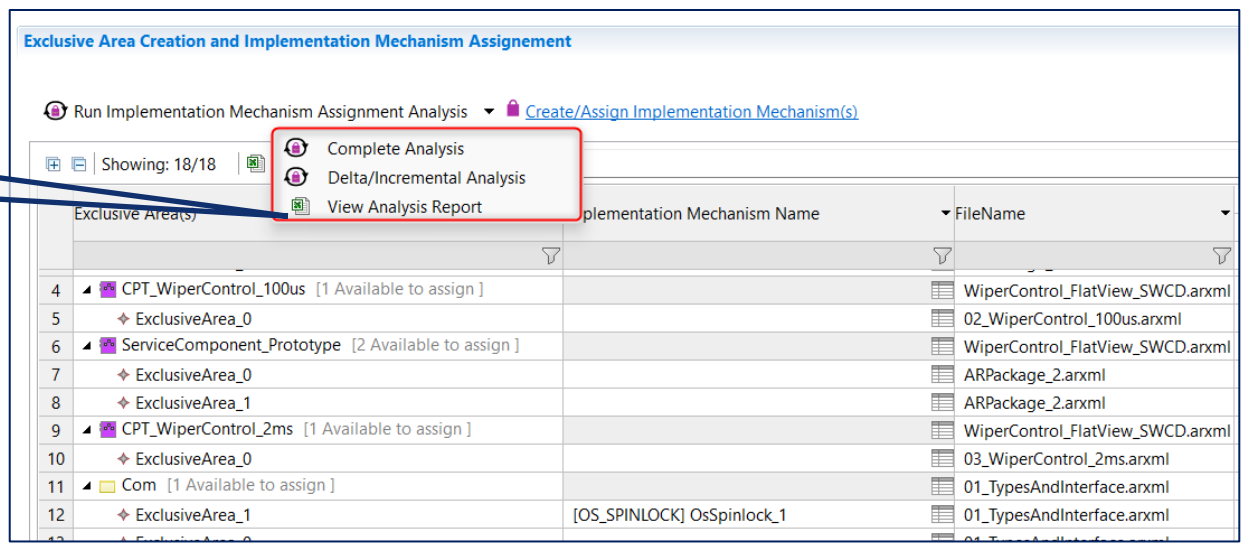
ISOLAR-A V12.7.0 What's New

Exclusive Area Lock Analysis(2/3)

Invocation

- Open RTE Editor > select tab “Exclusive Area and Impl Mechanism Config” > select “Run Implementation Mechanism Assignment Analysis” > Complete Analysis
- Delta/Incremental Analysis can be executed, when c or h files have changed after the first run of complete analysis.

Option to view generated report



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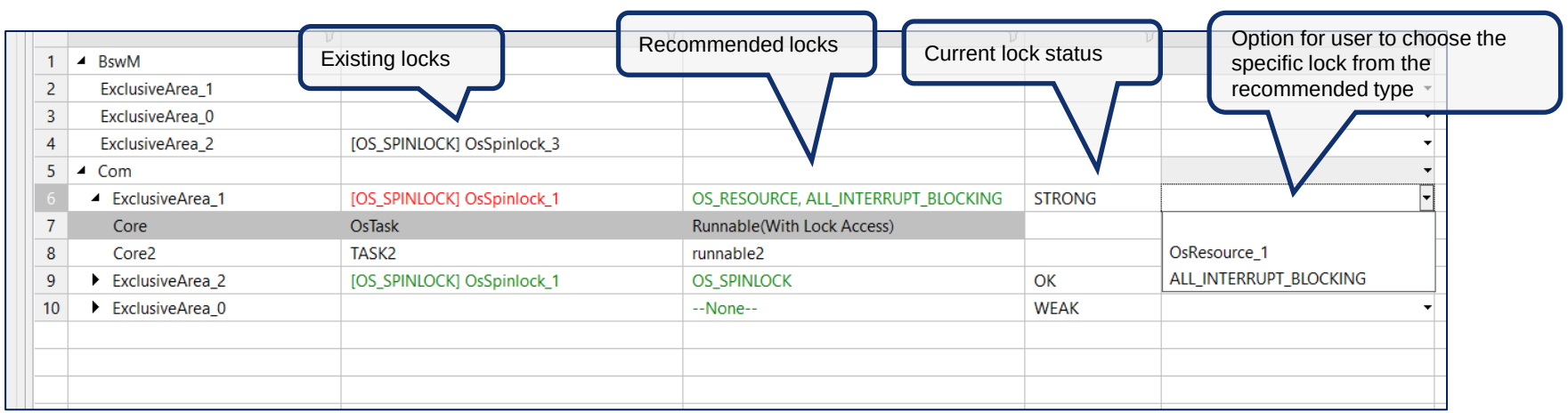
ISOLAR-A V12.7.0 What's New

Exclusive Area Lock Analysis (3/3)

Analysis Recommendation

- Once the analysis is complete, a recommendation dialog pops up displaying the lock recommendation along with current lock status.
- User can accept the recommendation by selecting the mechanism from the last column.
- A report gets generated once assignment is completed.



1	BswM				
2	ExclusiveArea_1				
3	ExclusiveArea_0				
4	ExclusiveArea_2	[OS_SPINLOCK] OsSpinlock_3			
5	Com				
6	ExclusiveArea_1	[OS_SPINLOCK] OsSpinlock_1	OS_RESOURCE, ALL_INTERRUPT_BLOCKING	STRONG	
7	Core	OsTask	Runnable(With Lock Access)		
8	Core2	TASK2	runnable2		
9	ExclusiveArea_2	[OS_SPINLOCK] OsSpinlock_1	OS_SPINLOCK	OK	
10	ExclusiveArea_0		--None--	WEAK	

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ISOLAR-A V12.7.0 What's New

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ISOLAR-A V12.7.0 What's New

Secured PDU support in DBC Import (1/2)

Background Information

- A message in DBC file can be protected as secured by using BA attribute SC_MESSAGE.
- Currently DBC Importer do not support secured messages and import it as normal message.

Feature Summary

- DBC importer is enhanced to process and import secured message as SecuredIPdu.

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ISOLAR-A V12.7.0 What's New

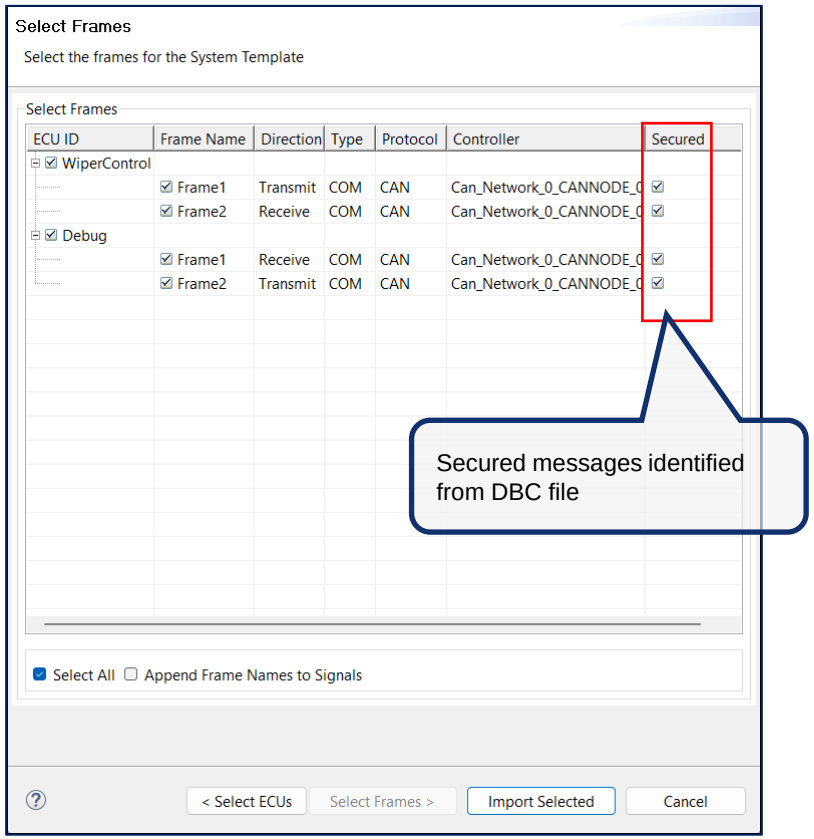
Secured PDU support in DBC Import (2/2)

Pre-requisites

– For a message to be secured, BA attribute “SC_MESSAGE” should be set to “Yes”

Custom parameter support

– Custom parameters SCAP_AuthInfoTxLength and SCFP_FreshnessValueTxLength are supported in BA_Def tags and will be imported into SecureCommunicationAuthenticationProps.AuthInfoTxLength and SecureCommunicationFreshnessProps.FreshnessValueTxLength, respectively.



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ISOLAR-A V12.7.0 What's New

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ISOLAR-A V12.7.0 What's New

Merge Mode Support in Java Validation (1/2)

Background Information

– Currently, by default user defined validation always run in a non-merged mode wherein user must take care of splitable handling through different APIs provided.

Feature Summary

– User defined validation is enhanced to run in merged mode by setting a flag in annotation.
– Legacy validations without this flag will run in non-merge mode (backward compatible).

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ISOLAR-A V12.7.0 What's New

Merge Mode Support in Java Validation (2/2)

- New Java validation files created through java action creation dialog will have the "enableMergeMode" parameter set to true by default, enabling merge mode.
- For backward compatibility, validations without "enableMergeMode", will run in unmerged mode.
- The execution mode (merged or unmerged) of user-defined validations depends on the annotation "enableMergeMode" value.
- If merge mode is enabled, then validation will be called only once per all split elements.

Annotation using which merge/unmerged mode can be controlled

```
@JavaActionValidation
public class ValidateMergeModeExecution implements IJavaValidation {

    @JavaActionValidation(profile = "Default", inputType = "CompositionSwComponentType",
            validationName = "validate", description = "", enableMergeMode = "true")
    public void validate(final IJavaValidationContext validationContext) {
        CompositionSwComponentType composition = validationContext.getInputContextElement();
        if (composition.getShortName().equals("WiperControl_SystemComposition")) {
            validationContext.getLog().error("No of connectors :" + composition.getConnectors
        }
```


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ISOLAR-A V12.7.0 What's New

File Level Validation Support for Custom Profile (1/2)

Background Information

– Currently, there is no option available to execute custom profile validation on file level in both GUI and CLI.

Feature Summary

– Option to invoke all profile validation including Custom Profile is provided on ARXML file level.
– CLI validation is improved to support Custom Profile & user defined validation along with other standard profile validations.

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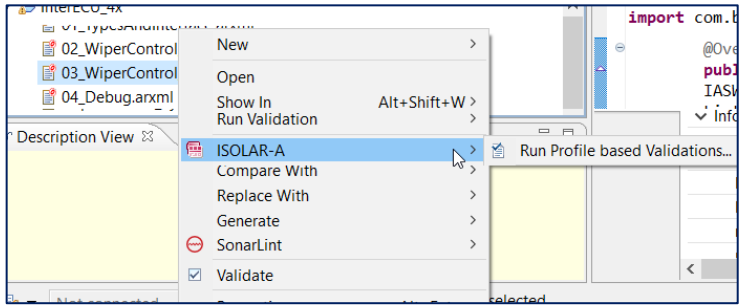
ISOLAR-A V12.7.0 What's New

File Level Validation Support for Custom Profile (2/2)

Invocation in GUI

– Select an ARXML file **>** ISOLAR-A **>** Run Profile Based Validations…



Invocation in CLI

– To execute custom profile at file level in CLI, the --project/-p argument is mandatory.

– Syntax: -arval --project=<Project Path> -f=<file path> --profile=<profile name>

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ISOLAR-A V12.7.0 What's New

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ISOLAR-A V12.7.0 What's New

General improvements (1/2)

Adapter Generation

– Init value(comspec) on source requestor parameter port is copied to generated provider port in the parameter component.

Component/SWC Editor

– DataReceivedEvent is configurable for NvData interface ports in both component and SWC editor.

AutoScheduling

– Scheduling based on OSTaskProxy is supported in CLI and Java Action

Command Line:
Syntax:

ISOLAR-AB.cmd -generate -autoschedule --type="ostaskproxy" –p=<project_path> --system= <System Name>

Example:

ISOLAR-AB.cmd -generate -autoschedule --type="ostaskproxy" –p="C:/temp/sample" --system= "System_ECUExtract"

Java Action:
Syntax:
boolean autoSchedule(String type, List<String> inputList, String system, additonalOptionalFile, logFile, ILog consoleLog)

Example:
autoSchedule("ostaskproxy", EMPTY_LIST, "System_ECUExtract","filepath.txt",inputContext.getLog());

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ISOLAR-A V12.7.0 What's New

General improvements (2/2)

Improve validation ASW4X_DT_0034

- Validation is extended to support SwServiceArg, SwPointerTargetProps and ImplementationDatatypeElements.

Report Generation Supports SecuredIPdu

- Report generation is enhanced fetch and include securedIPdu details in the generated report

UX Improvement in Custom Profile Dialog

- Default profiles are created on opening custom profile dialog.
- Validations included into a custom profile are removed from available list.

ECU Instance Splitable Property Support

- Below listed features are enhanced to support splitable property of Ecu Instance.
  - Swc To Ecu Mapping editor
  - DataMapping editor
  - Signal Router
  - ASW Autogen

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ISOLAR-B V12.7.0 What's New

Overview

1. **Improvements to RTA-CAR migrator summary page**

2. **Unification of ISOLAR-AB CLI***

3. **General Improvements**

* New feature in ISOLAR-B V12.7.0

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ISOLAR-B V12.7.0 What's New

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3. **General Improvements**

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ISOLAR-B V12.7.0 What’s New

Enhancements to project migrator summary page (1/2)

Background Information

- The project migrator summary page displays information regarding parameters which require manual intervention as part of migration.
- These entries only provide short name and fully qualified path information but does not provide any navigation support.

Feature Summary

- An interactive migrator summary page enables navigation to specific elements in the BSW Editor.
- The navigation support is extended for the same entries in Problems Log as well.

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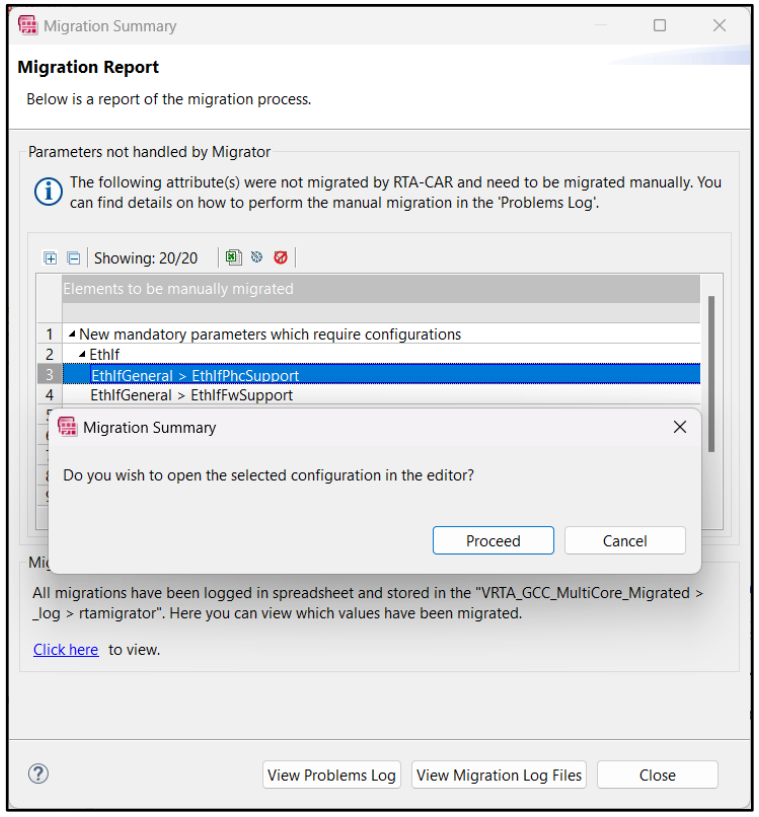
ISOLAR-B V12.7.0 What's New

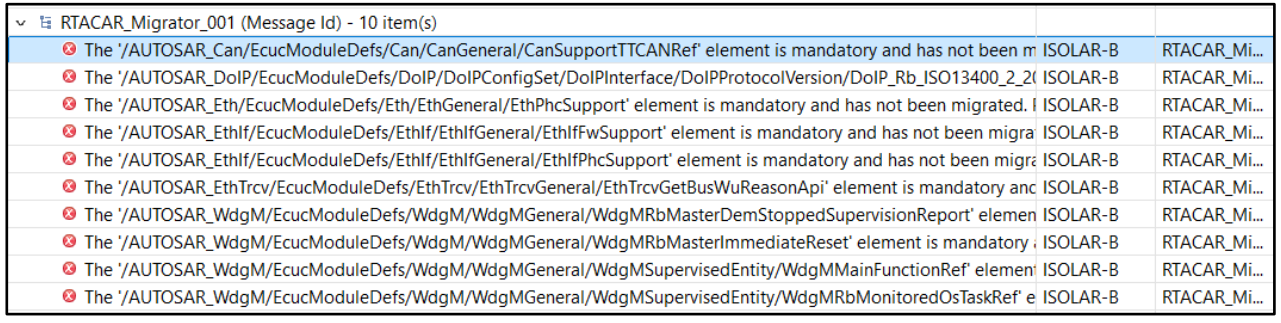
Enhancements to project migrator summary page (2/2)

Invocation

– Double click an entry > Proceed > Element opened in BSW Editor.

If the parameter configuration does not exist, its parent element is opened in the editor.





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ISOLAR-B V12.7.0 What's New

Overview

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ISOLAR-B V12.7.0 What’s New

Unification of command line interfaces in ISOLAR-AB (1/3)

Background Information

– The command line features of ISOLAR-A and ISOLAR-B can only be executed from their resp. command/shell script files.

Feature Summary

– A consistent and unified command line interface for ISOLAR-AB supporting single invocation point for all command line features of ISOLAR-A and ISOLAR-B (Backward compatible).
– Command line console provides ETAS Copyright and information about tool name and version along with improved error messages for incorrect and invalid commands.

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ISOLAR-B V12.7.0 What’s New

Unification of command line interfaces in ISOLAR-AB (2/3)

Invocation

– Open command prompt **>** "`ISOLAR-AB.cmd -h`"
  > ISOLAR-AB help contents displayed

– This will display information about both ISOLAR-A and ISOLAR-B commands together.

```
C:\I-AB\12.7.0_Mar18\ISOLAR-AB_V12.7.0-SNAPSHOT>ISOLAR-AB.cmd -h

C:\I-AB\12.7.0_Mar18\ISOLAR-AB_V12.7.0-SNAPSHOT>echo off
"ISOLAR-A Legacy Commands"
Commands available are:

        ISOLAR:
        ISOLAR-AB.cmd -adaptergen                                   ISOLAR-AB.cmd -arval
        ISOLAR-AB.cmd -aswautogen                                   ISOLAR-AB.cmd -autocleanup
        ISOLAR-AB.cmd -autoconfigurecomposition                     ISOLAR-AB.cmd -autoconnect
        ISOLAR-AB.cmd -autoportelementresourcemapping               ISOLAR-AB.cmd -autosignalmapping
        ISOLAR-AB.cmd -autoswc2ecumapping                           ISOLAR-AB.cmd -autoswc2implmapping
        ISOLAR-AB.cmd -clusterextract                               ISOLAR-AB.cmd -codeframe
        ISOLAR-AB.cmd -commatrix                                    ISOLAR-AB.cmd -deleteconnections
        ISOLAR-AB.cmd -ecuextract                                   ISOLAR-AB.cmd -ecupartition
        ISOLAR-AB.cmd -enablerips                                   ISOLAR-AB.cmd -executequery
        ISOLAR-AB.cmd -exportcomponent                              ISOLAR-AB.cmd -exportcomposition
        ISOLAR-AB.cmd -exportvariant                                ISOLAR-AB.cmd -importarxml
        ISOLAR-AB.cmd -importdbc                                    ISOLAR-AB.cmd -importfibex
        ISOLAR-AB.cmd -importldf                                    ISOLAR-AB.cmd -importodx
        ISOLAR-AB.cmd -isolarworkflow                               ISOLAR-AB.cmd -modelupdate
        ISOLAR-AB.cmd -reportgen                                    ISOLAR-AB.cmd -rttmimport
        ISOLAR-AB.cmd -runjava                                      ISOLAR-AB.cmd -runscript
        ISOLAR-AB.cmd -systemextract                                ISOLAR-AB.cmd -usernamingrules

For detailed help, please enter ISOLAR-AB.cmd command -h
Eg. ISOLAR-AB.cmd -gui -h

"ISOLAR-A Meta Commands"
Commands available are:

        ISOLAR-A META:
        ISOLAR-AB.cmd -cleanup -composition                         ISOLAR-AB.cmd -cleanup -system
        ISOLAR-AB.cmd -export -cluster                              ISOLAR-AB.cmd -export -commatrix
        ISOLAR-AB.cmd -export -component                            ISOLAR-AB.cmd -export -composition
        ISOLAR-AB.cmd -export -report                               ISOLAR-AB.cmd -export -system
        ISOLAR-AB.cmd -export -variantresolved                      ISOLAR-AB.cmd -generate -adapter
        ISOLAR-AB.cmd -generate -aswfromnw                          ISOLAR-AB.cmd -generate -autoportelementresourcemapping
        ISOLAR-AB.cmd -generate -autoschedule                       ISOLAR-AB.cmd -generate -codeframe
        ISOLAR-AB.cmd -generate -composition                        ISOLAR-AB.cmd -generate -connectors
        ISOLAR-AB.cmd -generate -ecuextract                         ISOLAR-AB.cmd -generate -ecupartition
        ISOLAR-AB.cmd -generate -enablerips                         ISOLAR-AB.cmd -generate -signalmapping
        ISOLAR-AB.cmd -generate -swc2ecumapping                     ISOLAR-AB.cmd -generate -swc2implmapping
        ISOLAR-AB.cmd -import -arxml                                ISOLAR-AB.cmd -import -dbc
        ISOLAR-AB.cmd -import -fibex                                ISOLAR-AB.cmd -import -ldf
        ISOLAR-AB.cmd -import -odx                                  ISOLAR-AB.cmd -import -rttm
        ISOLAR-AB.cmd -run -java                                    ISOLAR-AB.cmd -run -query
        ISOLAR-AB.cmd -run -script                                  ISOLAR-AB.cmd -run -usernamingrules
        ISOLAR-AB.cmd -run -workflow

For detailed help, please enter ISOLAR-AB.cmd command -h
Eg. ISOLAR-AB.cmd -gui -h

"ISOLAR-B Commands"
Commands available are:

        ISOLAR-B:
        ISOLAR-AB.cmd -arcompare                            ISOLAR-AB.cmd -bswModuleEnhancer
        ISOLAR-AB.cmd -bswbuild                             ISOLAR-AB.cmd -bswmoduleenhancer
        ISOLAR-AB.cmd -bswupdatemodel                       ISOLAR-AB.cmd -bswval
        ISOLAR-AB.cmd -build                                ISOLAR-AB.cmd -clean
        ISOLAR-AB.cmd -createProject                        ISOLAR-AB.cmd -ecuextractimport
        ISOLAR-AB.cmd -executequery                         ISOLAR-AB.cmd -generate
        ISOLAR-AB.cmd -jacoptest                            ISOLAR-AB.cmd -junittest
        ISOLAR-AB.cmd -migrate                              ISOLAR-AB.cmd -packageManager -export
        ISOLAR-AB.cmd -packageManager -install              ISOLAR-AB.cmd -projectmigrator
        ISOLAR-AB.cmd -rebuild                              ISOLAR-AB.cmd -script

For detailed help, please enter ISOLAR-AB.cmd command -h
Eg. ISOLAR-AB.cmd -gui -h
```


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ISOLAR-B V12.7.0 What’s New

Unification of command line interfaces in ISOLAR-AB (3/3)

Overview

– ETAS Copyright and tool name and version information shall be on every command execution.

```
C:\I-AB\ISOLAR-AB_V12.7.0-SNAPSHOT>ISOLAR-AB.cmd -bswval -p C:\ISolar-B\Tasks\ARCTOOLS-4111\JavaActions

C:\I-AB\ISOLAR-AB_V12.7.0-SNAPSHOT>echo off

##############################################################################
Copyright (c) ETAS GmbH 2025. All rights reserved.
##############################################################################
```

```
##############################################################################
Summary for -bswval command
Executed on 2025/03/18 18:54:53
ISOLAR-A/B 12.7.0b1 Standalone
##############################################################################
```

– Incase of invalid arguments or missing argument, user-friendly error messages are displayed.

```
C:\I-AB\ISOLAR-AB_V12.7.0-SNAPSHOT>ISOLAR-AB.cmd -adaptergenerate -h

C:\I-AB\ISOLAR-AB_V12.7.0-SNAPSHOT>echo off
The command '-adaptergenerate' is not valid. Kindly execute "ISOLAR-AB.cmd -h" to view the list of available commands in the tool.
```


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ISOLAR-B V12.7.0 What's New

Overview

1. Improvements to RTA-CAR migrator summary page

2. Unification of ISOLAR-AB CLI*

3. **General Improvements**

* New feature in ISOLAR-B V12.7.0

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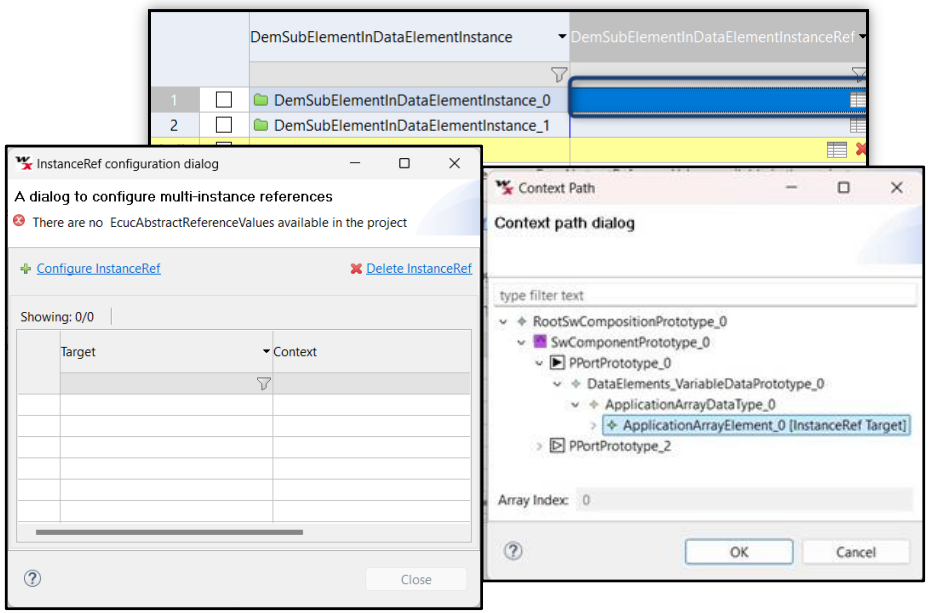
ISOLAR-B V12.7.0 What's New

General Improvements (1/1)

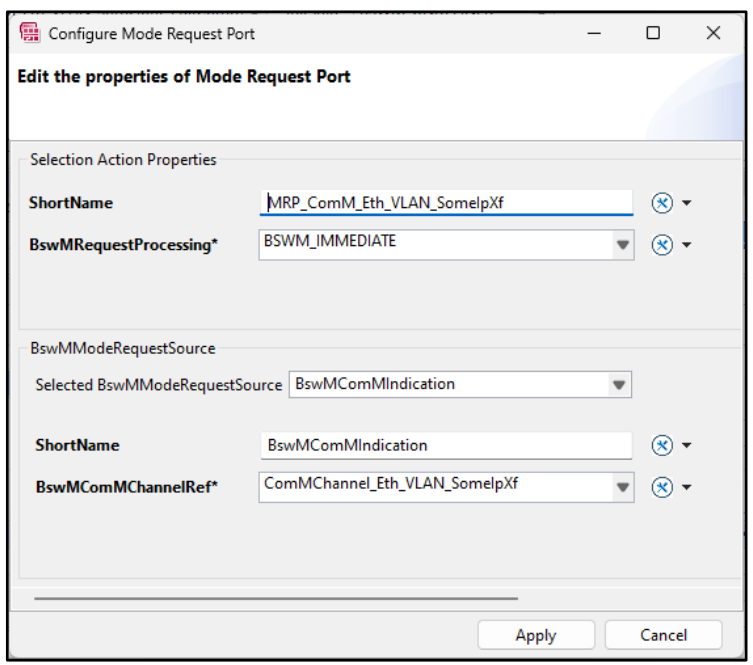
Enhancements to instance references in BSW Editor

– Configuration of instance references in table page.



Cancel support for BswM Editor attribute dialog

– Configuration changes can be discarded from the dialog without save.



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有難うございました　谢谢　감사합니다　Obrigado　Спасибо

Muchas gracias　Merci　धन्यवाद　Vielen Dank　Thank you